



University of Aden

Faculty of Science

Second Semester Exam. For The Academic Year 2018-2019

Department: Chemistry

Sitting Exam:

Level: 2nd

Specialization: Chemistry

Course: Gravimetric Analysis

Examiner: Dr. Adel Ahmed Mohammed Saeed

Time: 2 hours

Name of student:

PART I: ANSWER ALL THE QUESTIONS:

Name of student:

Q1. Circle the correct answer (2 marks each)

Gravimetric methods are based on measuring the mass of a pure compound to which the _____ is chemically related.

- (1) precipitate (2) mother (3) analyte (4) sample

In nucleation, about four or five ions, atoms, or molecules come together to form a stable _____ solid.

- (1) 100 nm diameter (2) 1-100 cm diameter (3) 1 cm diameter (4) 1 – 10 cm diameter

_____ is a separation technique that results in the formation of high pure solid particles from a solution.

- (1) Digestion (2) Extraction (3) Deposition (4) Crystallization

Among the anions, those which give insoluble precipitates with silver in an acidic solution will naturally interfere _____.

- (1) Br^- & CN^- (2) CO_3^{2-} & PO_4^{3-} (3) H^- & I^- (4) Br^- & $\text{C}_2\text{O}_4^{2-}$

_____ can reduce perchlorate, (ClO_4^-).

- (1) Zn^0 (2) FeSO_4 (3) $\text{Ti}_2(\text{SO}_4)_3$ (4) NH_4OH

Colloidal suspensions can often be coagulated by heating, _____, and adding an electrolyte.

- (1) diluting (2) stirring (3) igniting (4) solvating

_____ is called a Von Weimarn ratio.

- (1) $Q / (S - Q)$ (2) $(Q + S) / S$ (3) $S / (Q - S)$ (4) $(Q - S) / S$

Solubility is the _____ amount of the substance.

- (1) minimum (2) maximum (3) equal (4) fixed

To demonstrate the size of colloidal particles, we perform a _____ experiment.

- (1) decantation (2) flotation (3) dialysis (4) filtration

_____, the medium used in the reaction between silver and ascorbic acid.

- (1) H_2SO_4 (2) HCl (3) NH_4OH (4) HNO_3

Q.2 Fill in the blank (1 mark each)

Name of student:

Low relative supersaturation

☐ Less -----

☐ Fewer larger crystals

----- particles form

☐ Low-----

☐ Less ----- of -----



Some advantages of Gravimetric Analysis:

(1)

(2)

(3)

PART II: CHOOSE ONLY TWO (Calculator is Allowed) Name of student:

Q.3 Perform the following calculations (10 marks each)

- 1) Predict whether a precipitate will form if 30.0 mL of 0.010 mol/L silver nitrate is mixed with 25.0 mL of 0.0080 mol/L potassium chloride. ($K_{sp} = 1.8 \times 10^{-10}$)
- 2) An iron ore was analyzed by dissolving a 1.222-g sample in concentrated HCl. The resulting solution was diluted with water, and the iron(III) was precipitated as the hydrous oxide $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ by the addition of NH_3 . After filtration and washing, the residue was ignited at a high temperature to give 0.4194 g of pure Fe_2O_3 (159.69 g/mol). Calculate (a) the % Fe (55.847 g/mol) and (b) the % Fe_3O_4 (231.54 g/mol) in the sample.
- 3) A sample of blue copper(II) sulfate crystals weighing 2.55 g is heated and decomposes to produce 1.63 g of anhydrous copper(II) sulfate. Show that the formula of the blue crystals is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. (M.wt.: $\text{CuSO}_4 = 159.5 \text{ g.mol}^{-1}$, $\text{H}_2\text{O} = 18.0 \text{ g.mol}^{-1}$)
- 4) A 523.1-mg sample of impure KBr is treated with excess AgNO_3 and 814.5 mg AgBr is obtained. What is the purity of the KBr? (A.wt.: K=39.098, Ag=107.86, Br=79.904)
- 5) 6.00 g of helium gas was blown into a fairground balloon. On the day, the temperature was 301 K and the pressure inside the balloon was 103.4 kPa. Assuming it is infinitely elastic, to what volume did the balloon inflate? (A.wt.: He = 4 g.mol⁻¹, R=8.31)

Name of student: